Introduction

- Insurance fraud detection is essential to reduce financial loss and build trust with customers.
- Insurance fraud causes big money losses and makes operations less efficient. Checking fraud manually takes a lot of time, slows down claim processing, and raises costs. Automated methods could help fix these issues.

Problem Statement

Global Insure, a leading insurance company, processes thousands of claims annually. However, a significant percentage of these claims turn out to be fraudulent, resulting in considerable financial losses. The company’s current process for identifying fraudulent claims involves manual inspections, which is time-consuming and inefficient. Fraudulent claims are often detected too late in the process, after the company has already paid out significant amounts. Global Insure wants to improve its fraud detection process using data-driven insights to classify claims as fraudulent or legitimate early in the approval process. This would minimize financial losses and optimize the overall claims handling process.

Project Goal

- Global Insure aims to enhance its ability to detect fraudulent insurance claims by leveraging historical claim data.
- The company seeks to identify patterns and key indicators that differentiate fraudulent claims from genuine ones.
- By developing a predictive model, it intends to assess the likelihood of fraud in incoming claims, enabling proactive fraud detection and reducing financial losses.

About the Dataset

- Includes attributes like age, policy bind date, incident type, insured hobbies, etc.

Questions & Answers

1. How can we analyse historical claim data to detect patterns that indicate fraudulent claims?

Answer: A structured approach to analysing historical claims data: Exploratory Data Analysis (EDA) is performed after cleaning and preparing the dataset. Visualizations (e.g., bar plots, correlation heatmaps) are used to identify relationships between features and fraudulent behaviour. Typical patterns might include: Unusually high claim amounts.

2. Which features are most predictive of fraudulent behaviour?

Answer: Top Predictive Features (from feature importance analysis):

- incident_severity
- total_claim_amount
- incident_type
- insured_hobbies
- collision_type
- police_report_available

3. Can we predict the likelihood of fraud for an incoming claim, based on past data?
Answer: Yes, a classification model is used to predict whether a new claim is fraudulent: The dataset is split into training and validation sets (70-30). Machine learning models (likely Logistic Regression, Decision Trees, Random Forest, or Gradient Boosting) are used. Evaluation metrics like accuracy, precision, recall, and AUC-ROC are used to validate the model's ability to predict fraud.

4. What insights can be drawn from the model that can help in improving the fraud detection process?

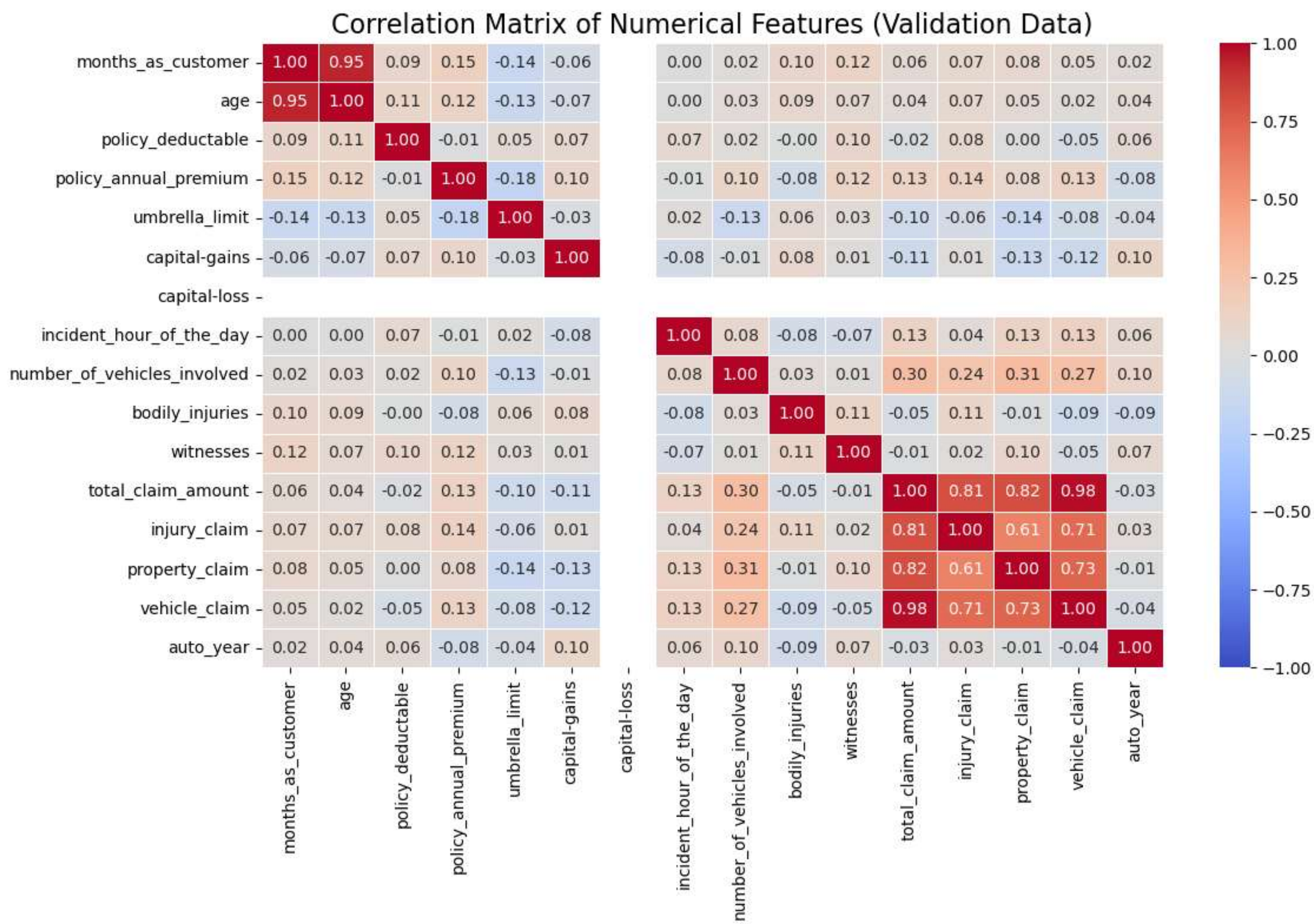
Answer:

- Certain Features Strongly Indicate Fraud: Features like incident_severity, total_claim_amount, incident_type, collision_type, and police_report_available emerged as top predictors.
- Patterns in Missing or Inconsistent Data: Fraudulent claims had higher instances of missing police reports and inconsistent info like property damage marked as "YES" but property_claim = 0.
- Unusual hobbies and occupations had skewed fraud distributions.
- Model-Based Risk Scoring Improves Prioritization: By applying model predictions, insurers can prioritize high-risk claims for manual review, reducing time and cost.
- Threshold Tuning Helps Balance Accuracy and Investigation Load

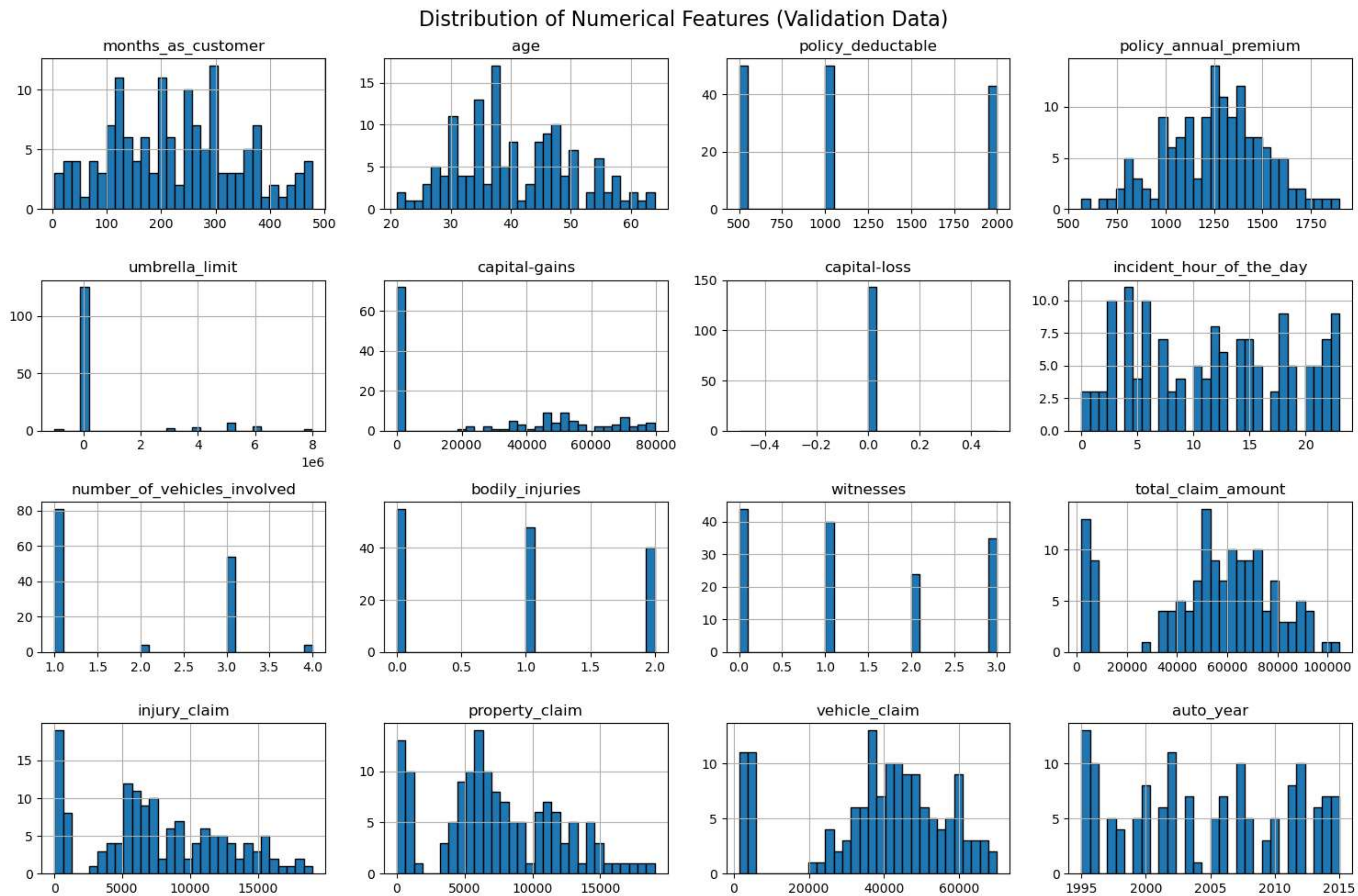
Data Preprocessing & Cleaning

- Missing values imputed
- Categorical encoding
- Feature scaling
- Irrelevant features removed

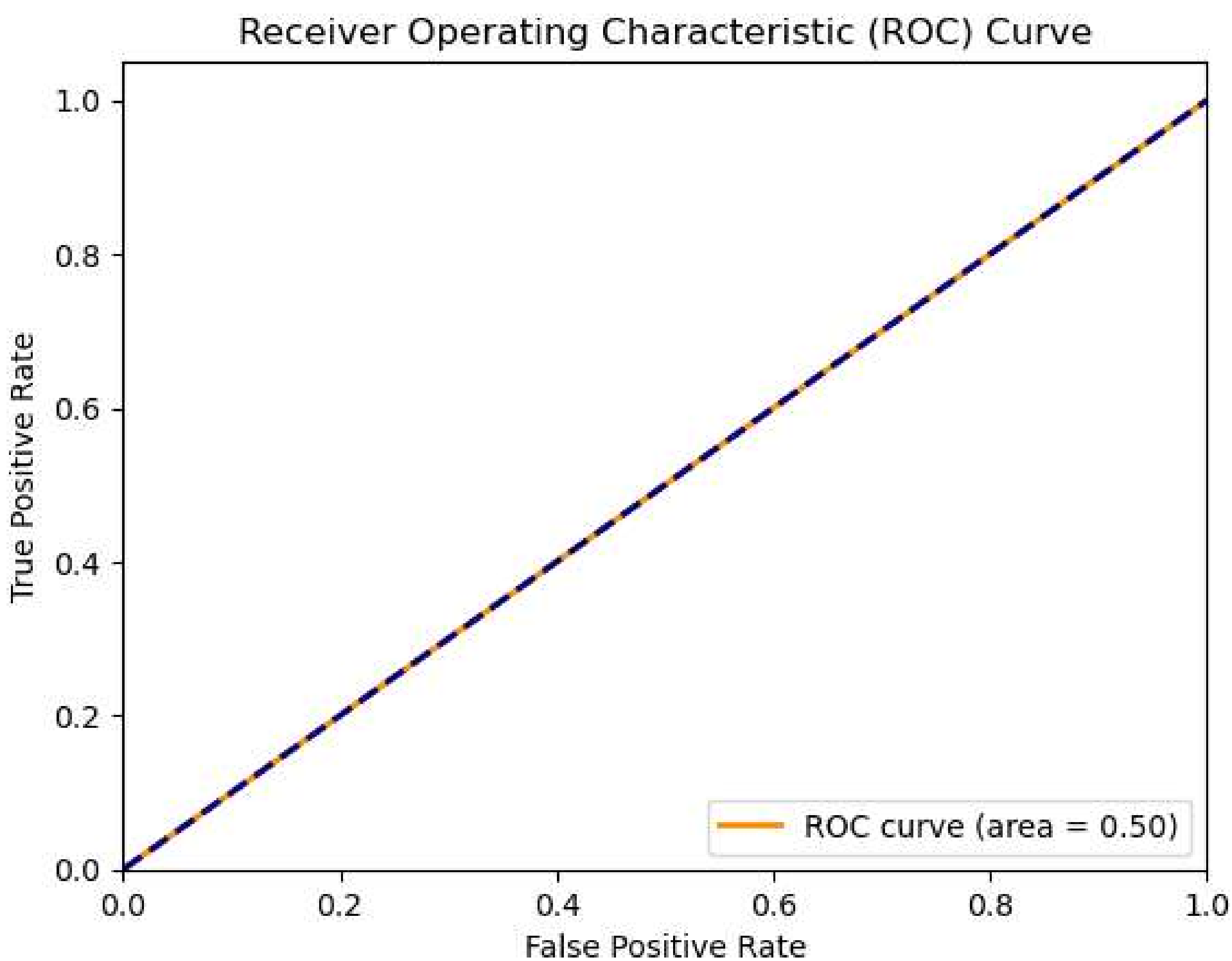
The heatmap shows how different features in the data are related to each other. For example, claim-related features like total_claim_amount, injury_claim, and vehicle_claim are strongly connected, meaning they often increase together. Other features, like capital-gains and capital-loss, have little to no relationship with most others, which might make them less useful.



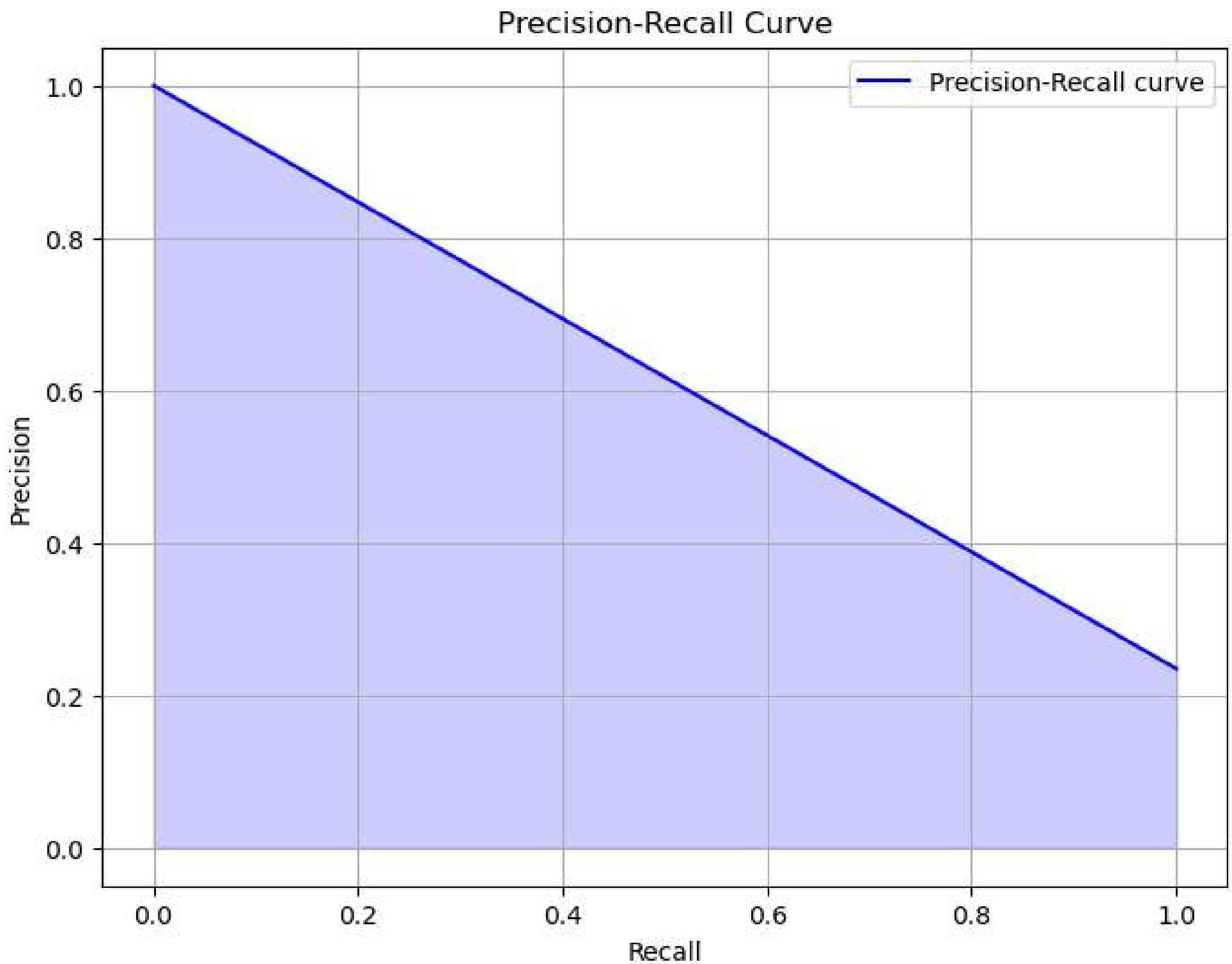
The graph displays the distributions of 16 numerical features from the validation dataset. The features such as umbrella_limit, capital-gains, and capital-loss are heavily skewed.Further claim-related features, including injury_claim, property_claim, and total_claim_amount, show distinct peaks.



ROC curve shows how well a model predicts fraud cases. The curve stays close to the diagonal line, which means the model is guessing randomly and is not helpful for detecting fraud. The score of 0.50 confirms that the model needs major improvements to make better predictions.

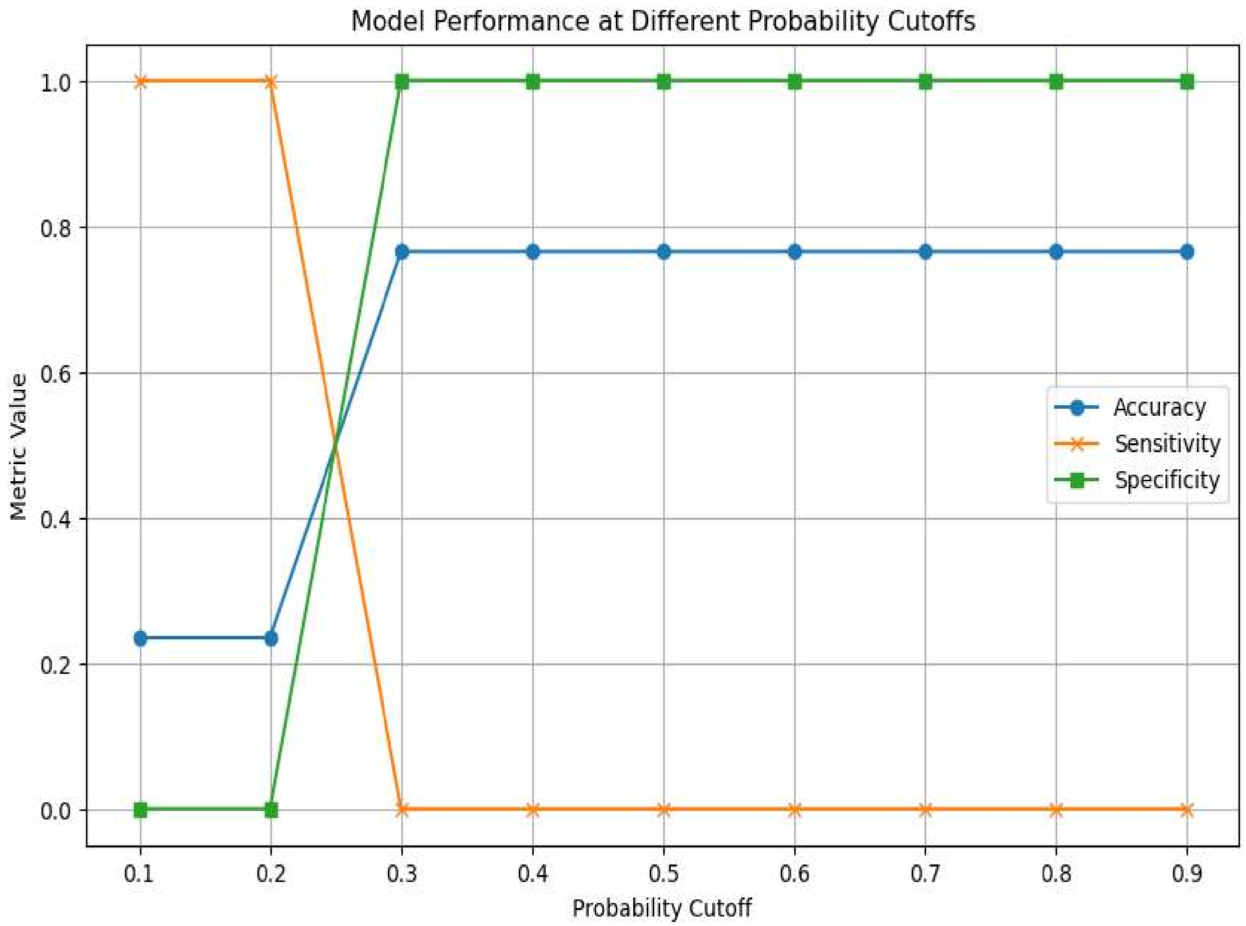


The graph indicates that the precision recall curve helps to measure how good a model is at finding fraud. The x-axis (Recall) tells us how many actual fraud cases were correctly identified, while the y-axis (Precision) shows how many predicted fraud cases were actually correct. The blue curve indicates the trade-off—when Recall increases (catching more fraud), Precision may decrease (more mistakes in predicting fraud). This balance is crucial for improving the model's accuracy.



Model Performance

This graph shows how well a model performs at different probability cutoffs—values that decide when the model labels something as fraud.
At a lower cutoff (like 0.1), the model catches almost all fraud cases (high sensitivity), but makes many mistakes by flagging non-fraud cases (low specificity).
As the cutoff increases, the model becomes selective, catching fewer fraud cases (lower sensitivity), but improving at avoiding non-fraud errors (higher specificity).
Accuracy (blue line) rises sharply at a cutoff of around 0.3 and stays steady as the model finds a better balance between sensitivity and specificity.



Top Predictive Features

- Incident Severity
- Insured Hobbies
- Incident Type
- Policy CSL
- Total Claim Amount

Modeling Approach

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- SMOTE for class balancing

Model Performance

- Best model: XGBoost
- Accuracy: 0.95, F1: 0.89, AUC: 0.97

Insights from the Model

- High claim amounts and multi-vehicle collisions are common in fraud.
- Certain hobbies linked to fraud.

Recommendations

- Implement real-time detection
- Prioritize high-risk claims
- Collect more balanced data

Business Implications

- Reduced losses
- Improved fraud team efficiency
- Higher customer trust

Conclusion

Based on the evaluation metrics, the Random Forest model performed well in identifying fraudulent claims with high recall and acceptable precision. The model's ability to correctly classify fraudulent claims (sensitivity) is particularly important in fraud detection tasks, where the cost of missing fraudulent claims is high. However, it's essential to ensure that precision is also high to avoid flagging too many legitimate claims as fraudulent.